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A close-up, blue-tinted image of a microchip. A white padlock icon is superimposed on the chip, with glowing blue lines radiating from it, symbolizing security and trust in hardware.

Simplify your security journey with STM32Trust TEE Secure Manager

Fabrice GENDREAU

Addressing the Security Challenges & Gaps



Security challenges
for our customers

Complex

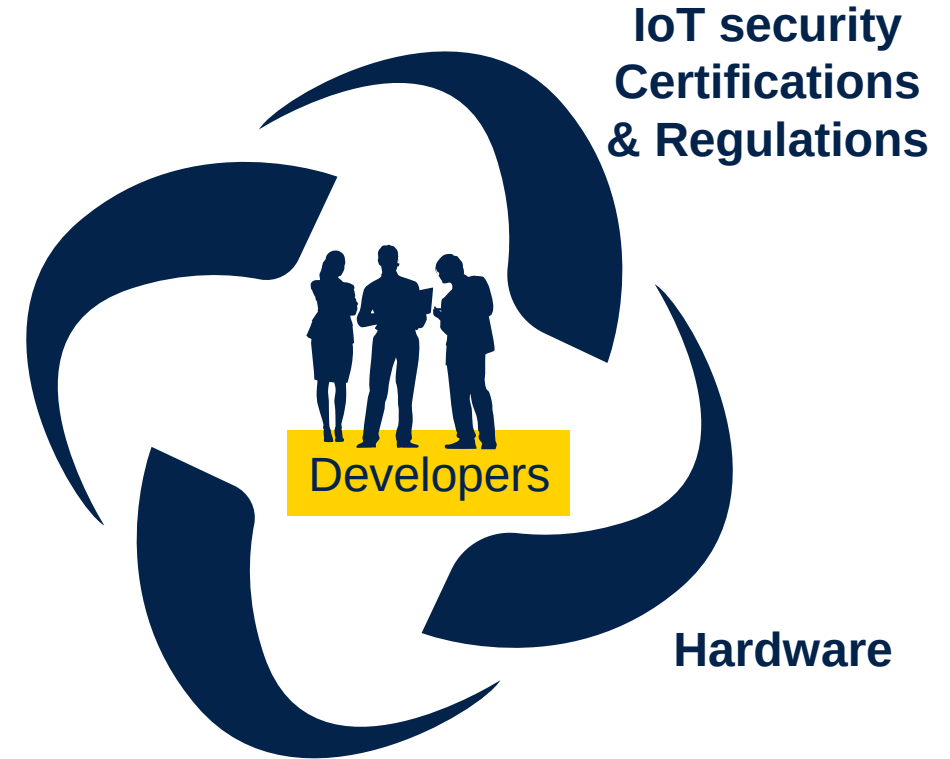
Evolving
fast

Time to
market

The missing link

Scalability, certification, maintenance
core security hardware and services

Multiple
Devices



IoT security Certifications & Regulations

Radio Equipment Directive (RED)

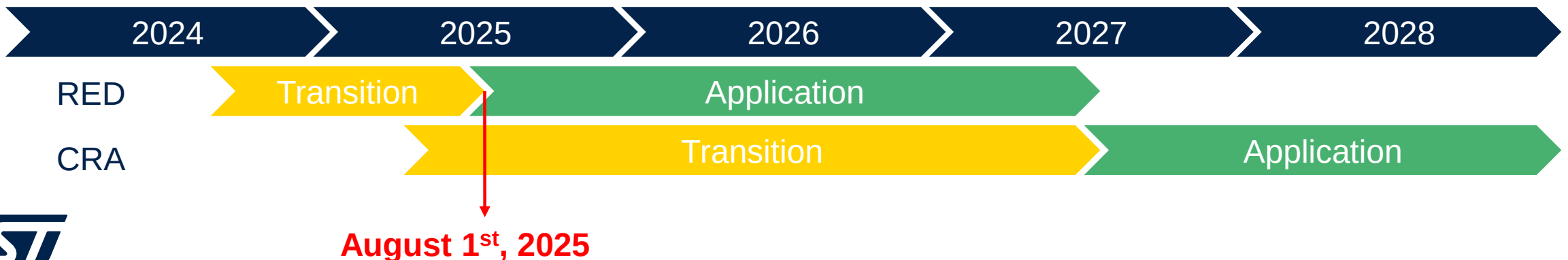
Goal: Essential requirements for **radio equipment**

- EMC, safety/health, **privacy** & **fraud protection**
- Be capable to update/patch the products.
- **Conformity assessment** with **risk-based approach** according to the usage and environment of the device.
 - HW Component : N/A
 - IoT Consumer & Industrial Device : Self-Declaration
 - Medical & Auto devices excluded

Cyber Resilience Act (CRA)

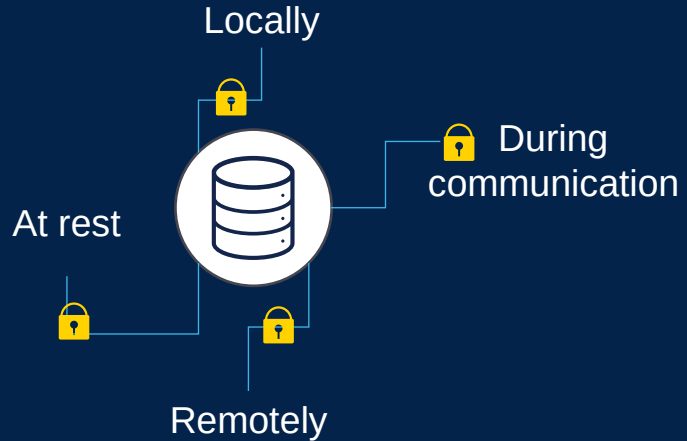
Goal: Ensures more secure **HW & SW products** on the market.

- **Actively monitoring** vulnerabilities and provide update/patch for the material.
- Different Security levels according to **pre-defined categories**.
 - HW Component : **Third party evaluation**
 - IoT Consumer Device : Self-Declaration
 - IoT Industrial Device : **Third party evaluation**

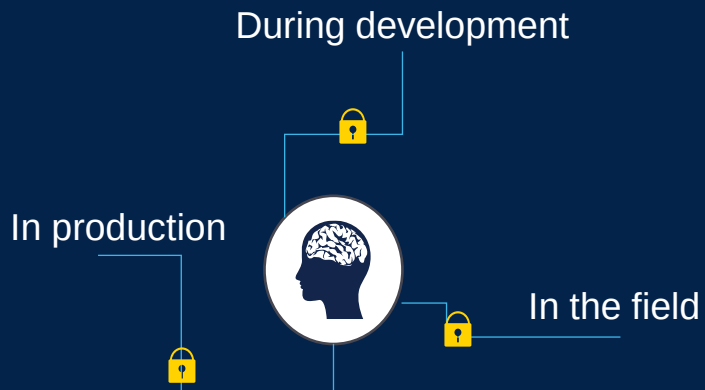


“ If only

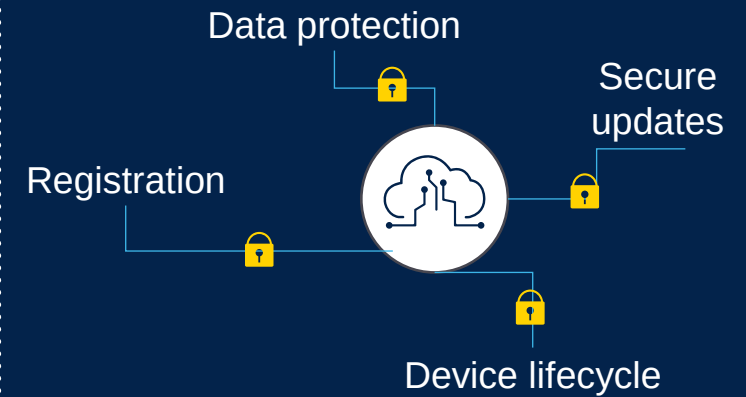
I could **easily** protect my critical **data & secrets** and those of my end customers



I could **easily** and **strongly** protect my **IPs**, and my partner's **IPs**



I could **easily** & **securely** connect to **Clouds & Servers** without Painful digital identities management



This is where we come in !

Secure Manager

A Trusted Execution Environment (TEE) integrating core security services

A turnkey set of security services developed, maintained and certified by ST

Security strategy



OUR VISION

Be the reference for securing
GP MCUs & MPUs

OUR MISSION

Provide our developers the
means to build
Secure applications

OUR STRATEGY



Security Robustness &
Assurance

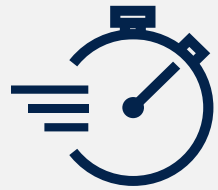


← Portfolio consistency →

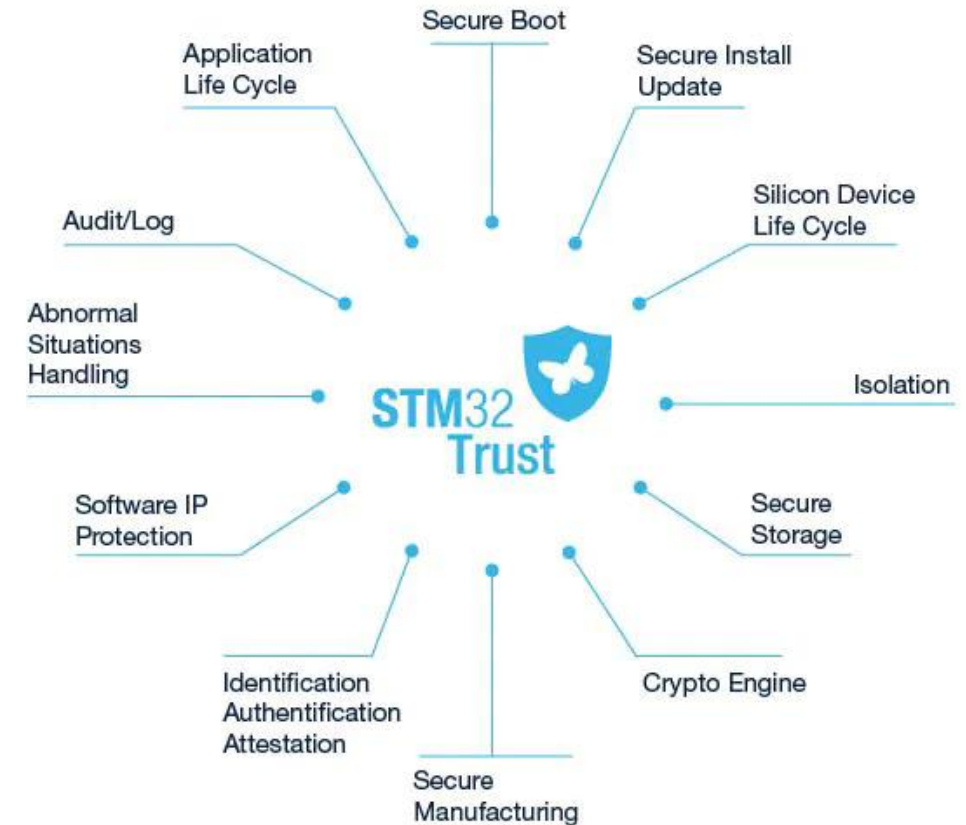
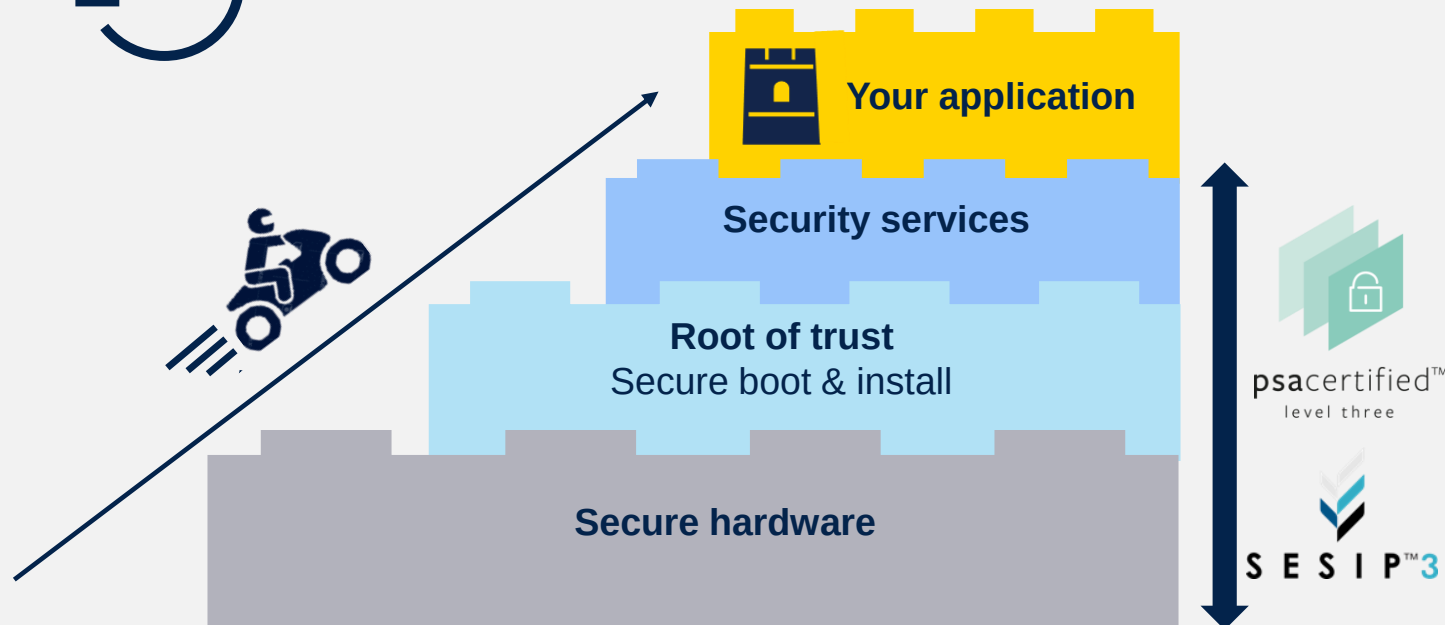


Scalable & Turn-key
solutions

Building Trust in Embedded Systems



Innovate faster!



STM32Trust TEE – Secure Manager



Scalable security to accelerate time to market

Secure Manager

A Trusted Execution Environment (TEE) integrating core security services

A simplified customer journey

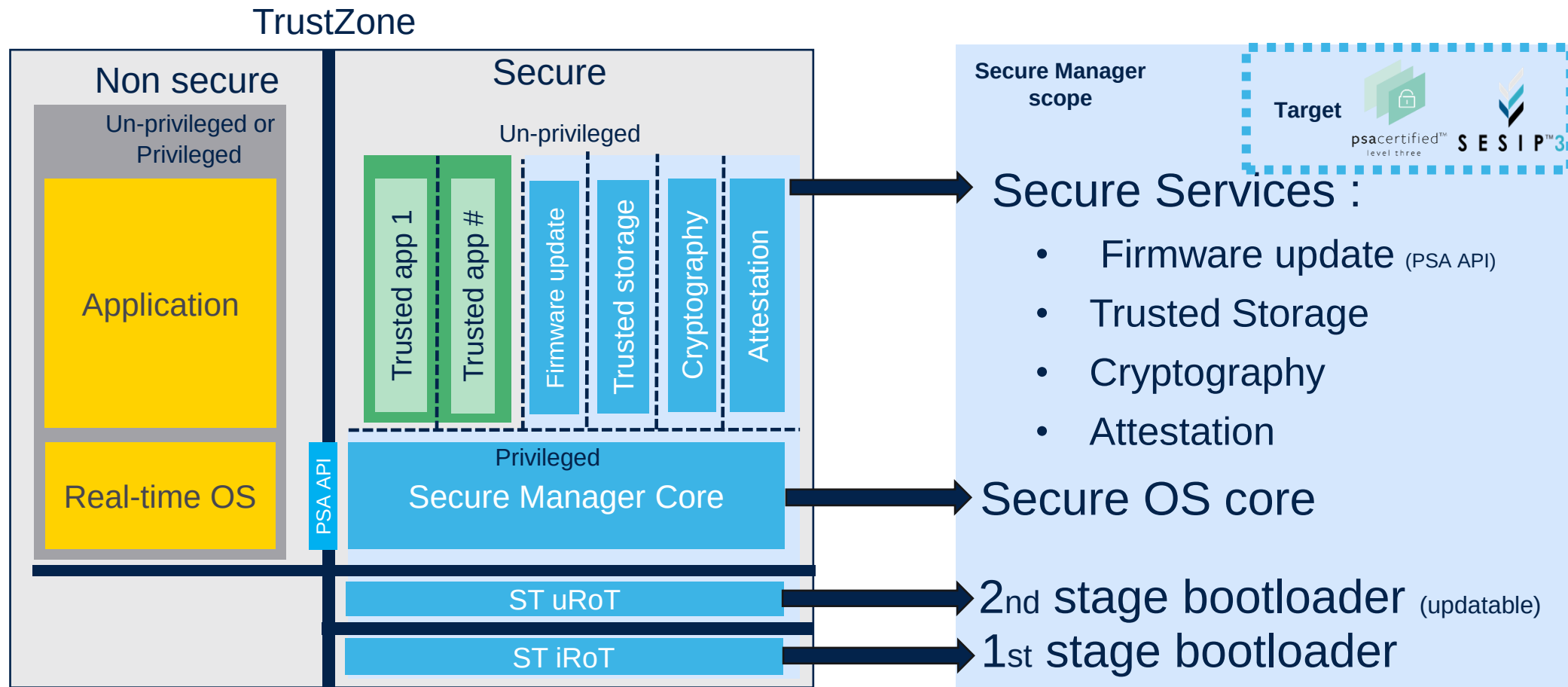
Seamless cloud/server support

Supporting remote provisioning

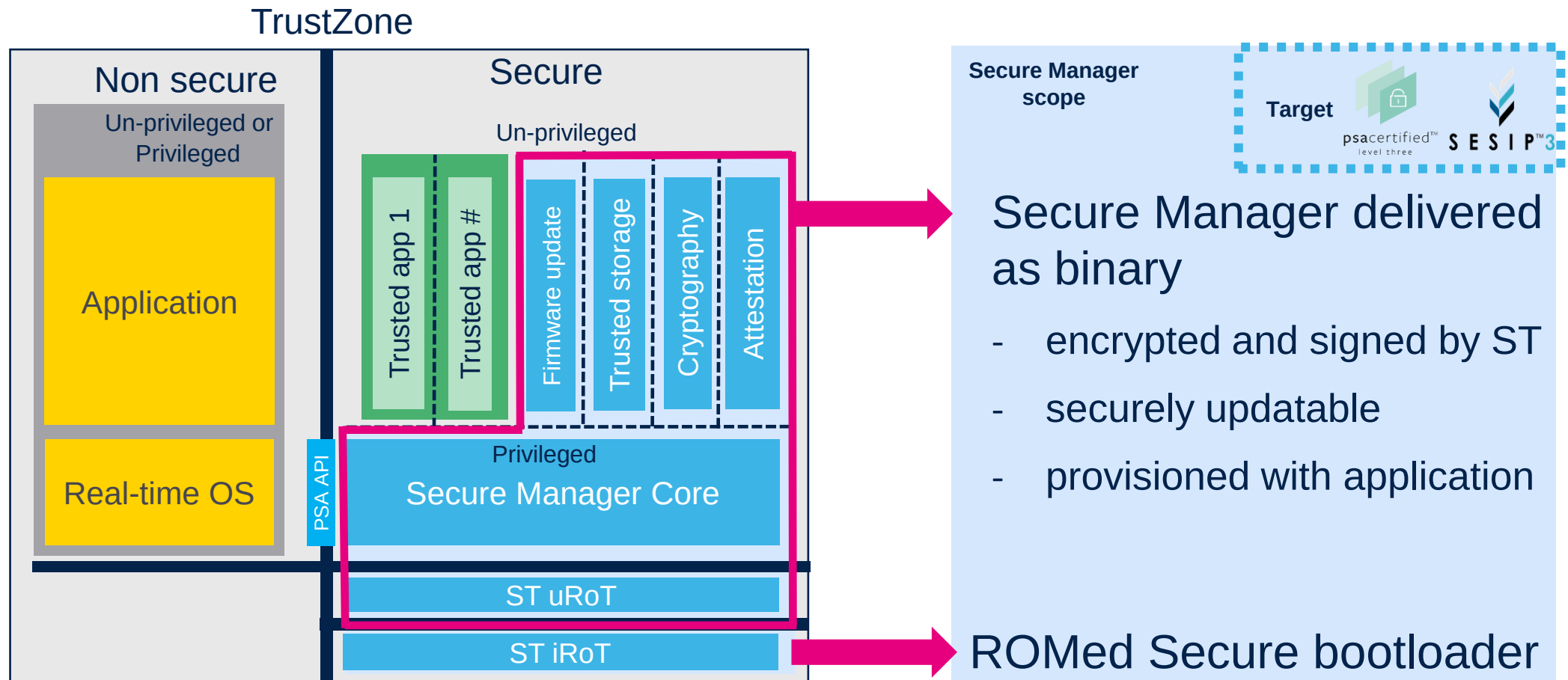
Multi-tenant IP protection

The first MCU supplier to offer a certified and maintained TEE solution to customers

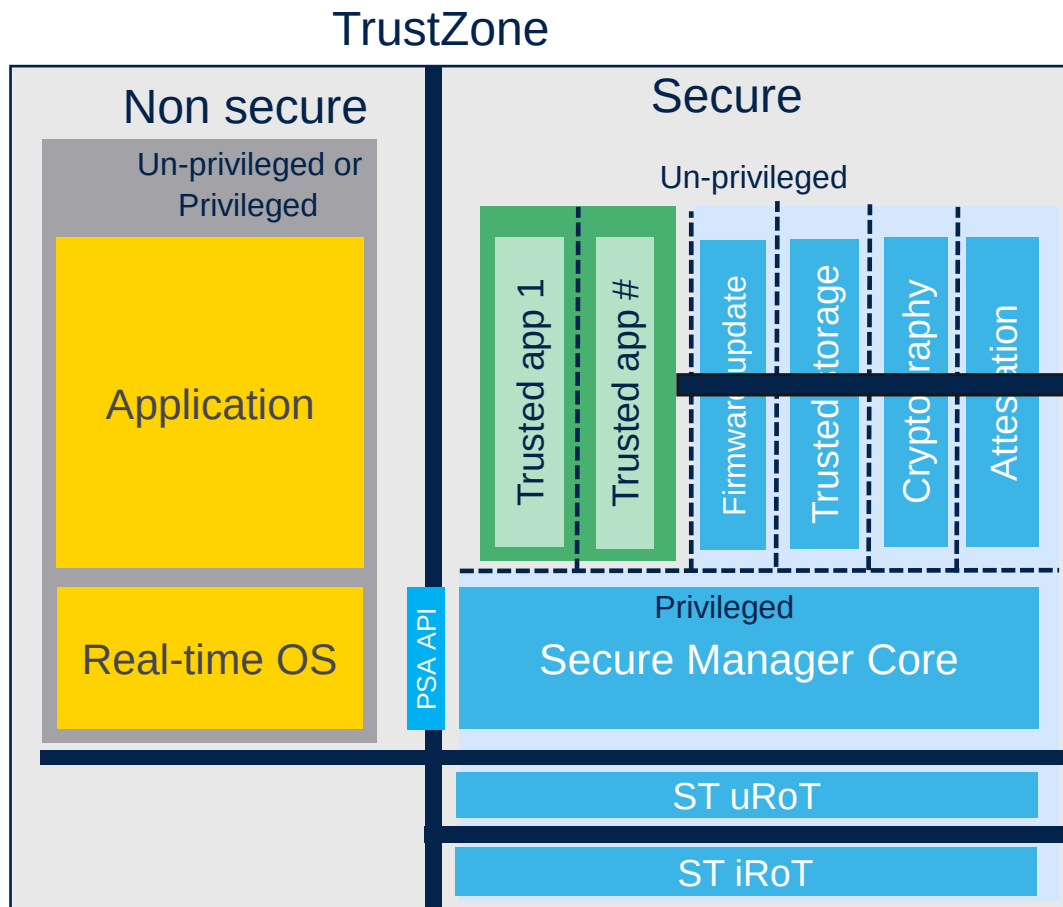
Protect IP and **simplify** security customer journey



Protect IP and **simplify** security customer journey



Protect IP and **simplify** security customer journey

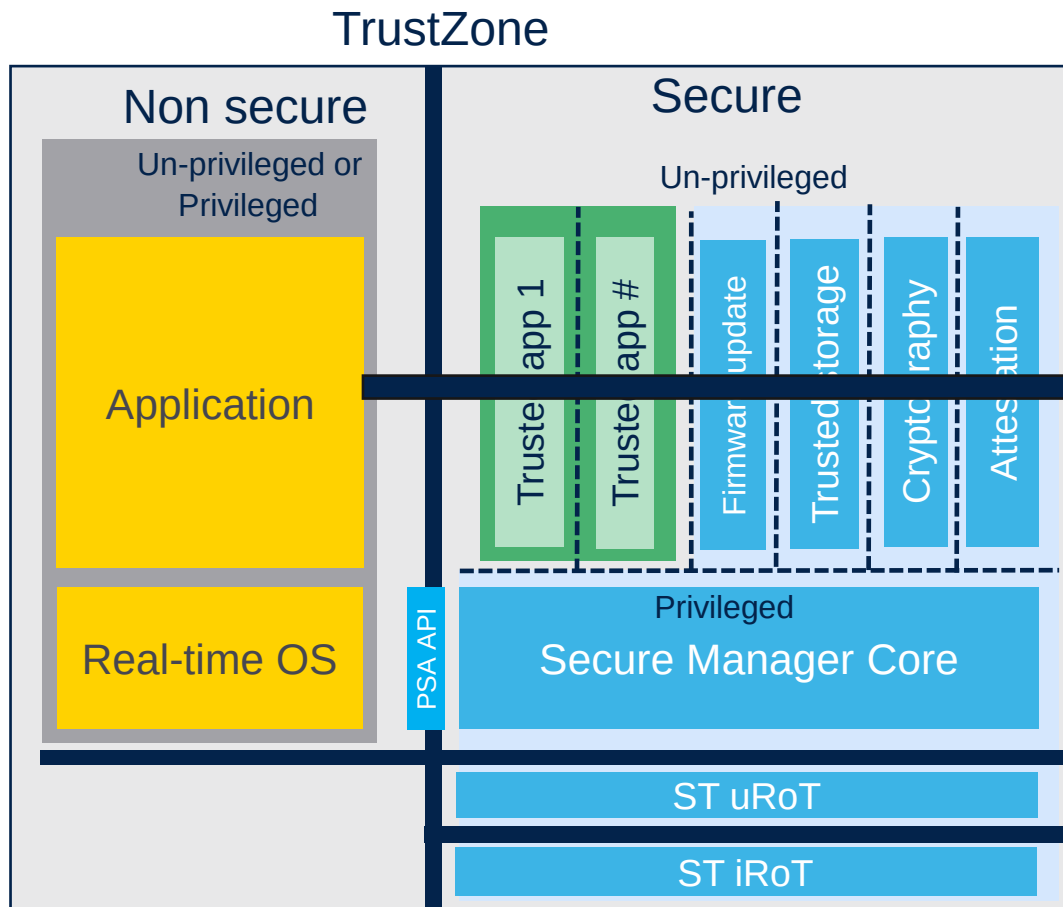


Third Party /
OEM scope

Secure Modules:
delivered as binary

- encrypted and signed by OEM/3rd party
- securely & independently updatable

Protect IP and **simplify** security customer journey

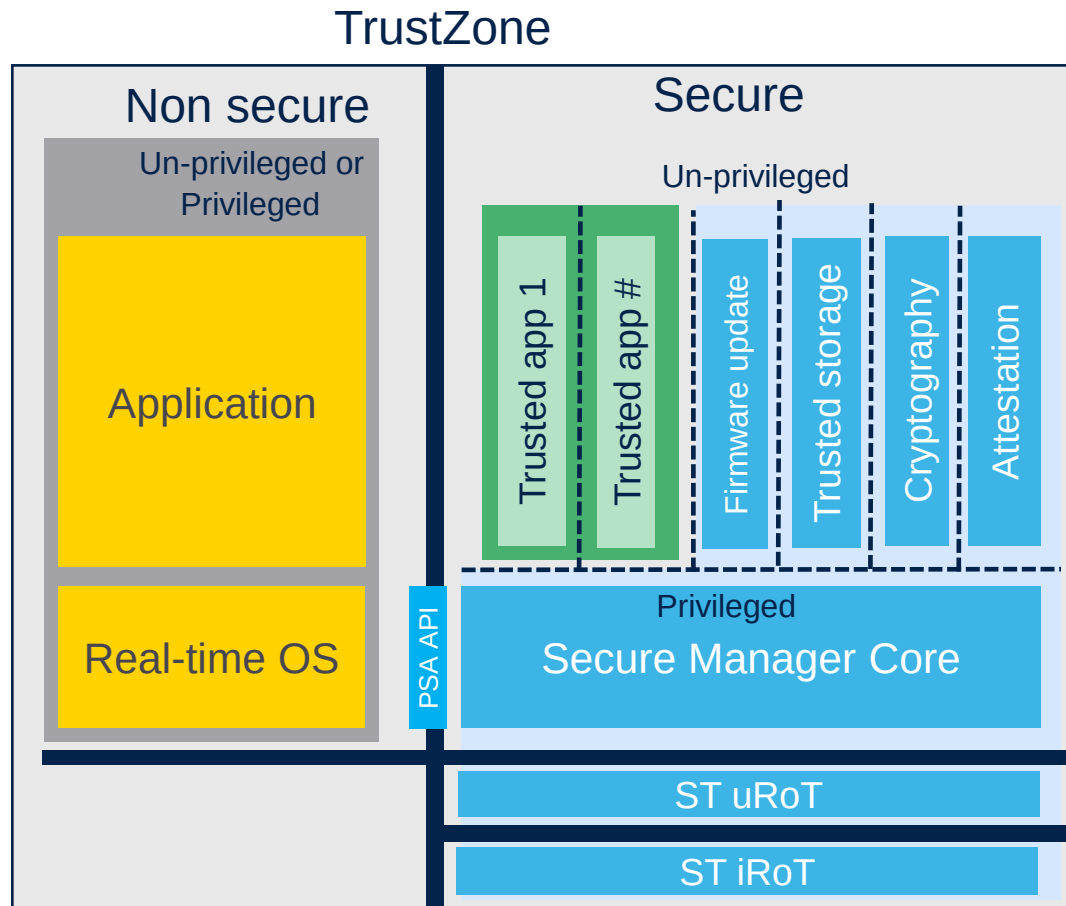


OEM scope

Non-Secure Application

- securely & independently updatable
- optionally encrypted

Protect IP and **simplify** security customer journey



- ST platform ownership
- Turnkey set of security services
- Secure Manager Core to handle isolation
- Multi-tenant software IP protection
- Arm® PSA API compatible
- Designed for Long-Term-Support
- Modular secure update capable
- Optimized certification properties
- Certified and maintained by ST



Scope of
Secure Manager

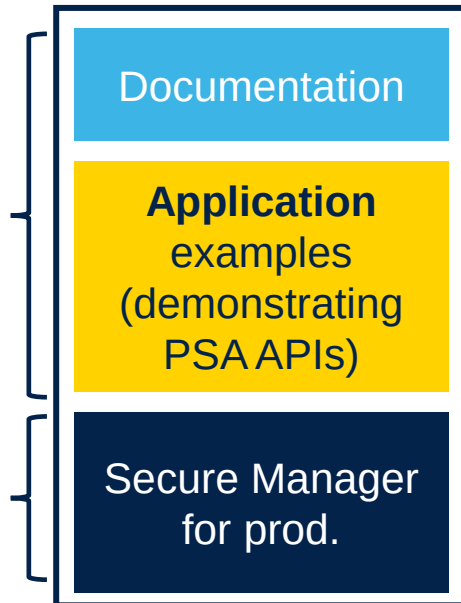
Simplify developer's experience

SMAK

To develop applications using security services

Downloaded from
[STM32CubeH5](#)
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Downloaded from
[STM32TRUSTEE-SM](#)
(encrypted binary)
license [SLA0044](#)

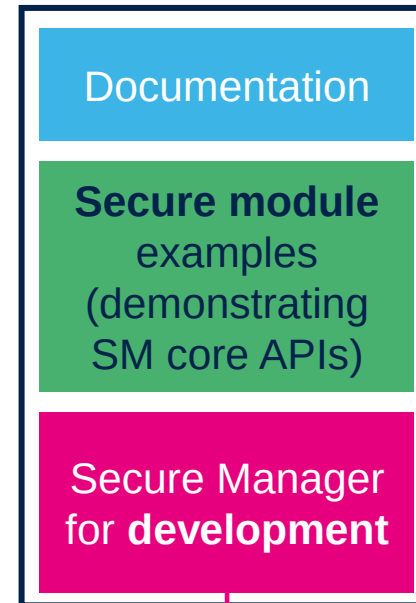


SMDK

To develop module inside TrustZone®

X-CUBE-SMDK-H5
Available on demand
(encrypted binary)

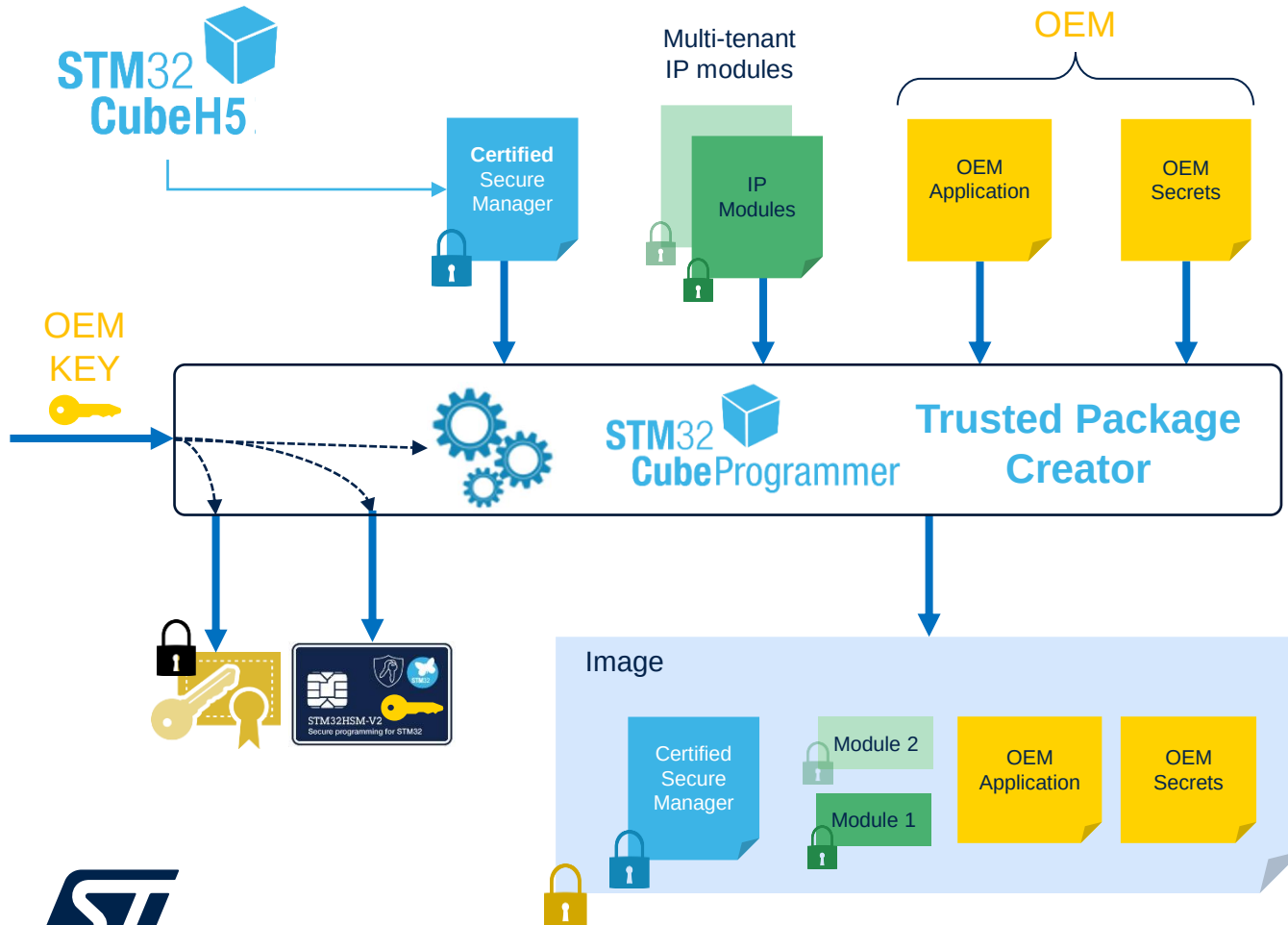
Signed LLA



Only for development

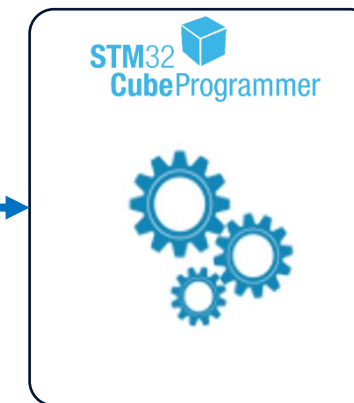
Secure Manager Development & Installation flow

Application Creation Flow

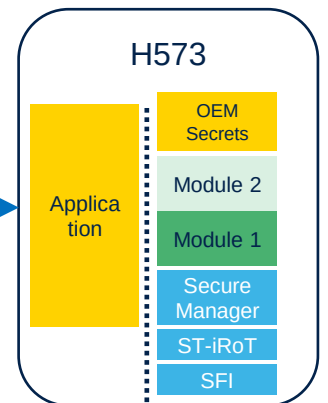


Application Installation Flow

Initial/virgin state



OEM personalized product



Secure Manager

Key messages

A background image of an industrial robotic arm in a factory setting, with various mechanical components and cables visible.

A certified turnkey SoC set of security services

In-ST-factory pre-provisioned identities for Clouds & OEM servers

An exhaustive software IP protection solution

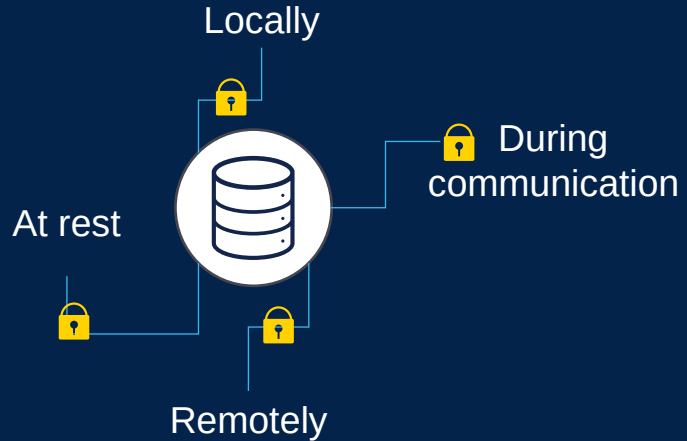
Enabling 3rd party keys & certificates Lifecycle management

Conclusion & Takeaways

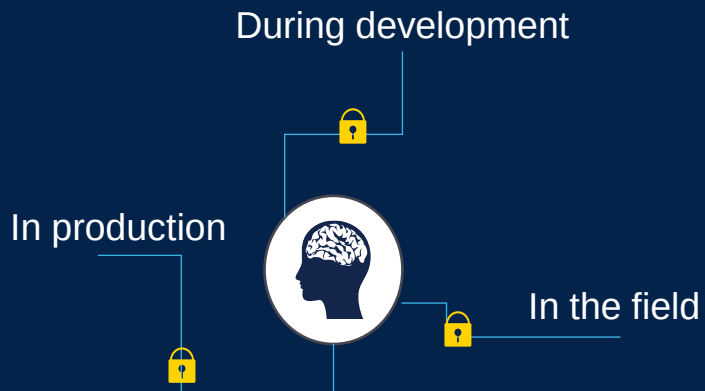


“ If only

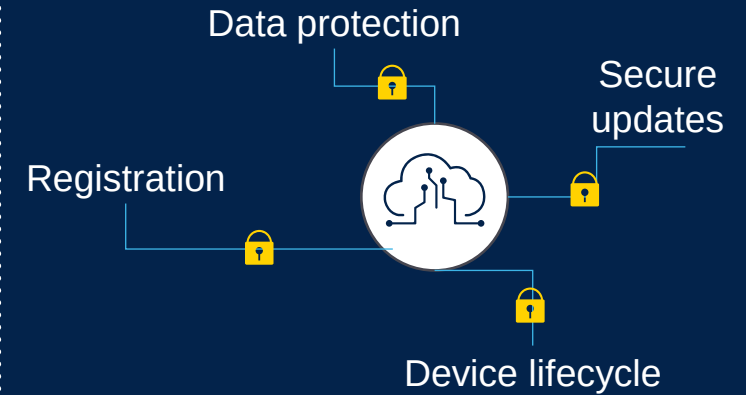
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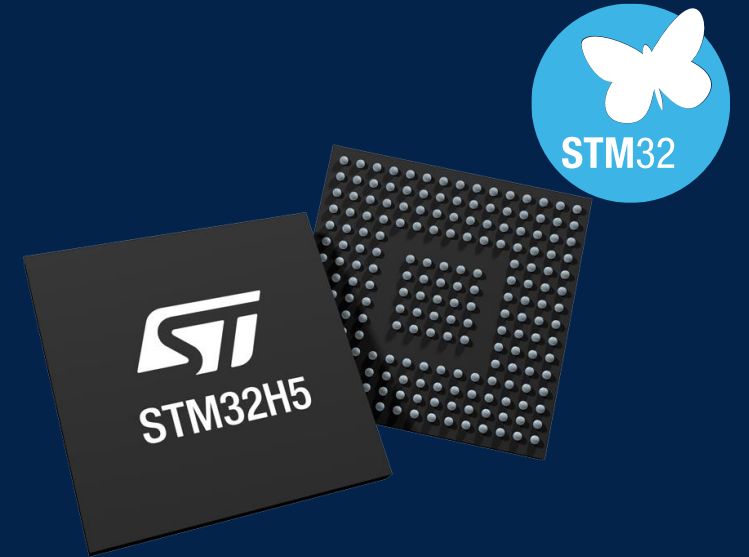


Scalable security

From **secure hardware**
to **full solution** owned &
maintained by ST

STM32
Trust  TEE

Secure Manager

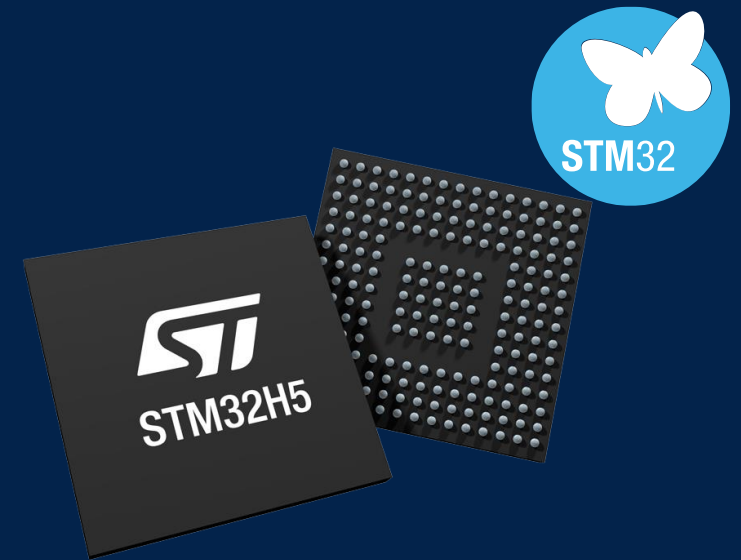
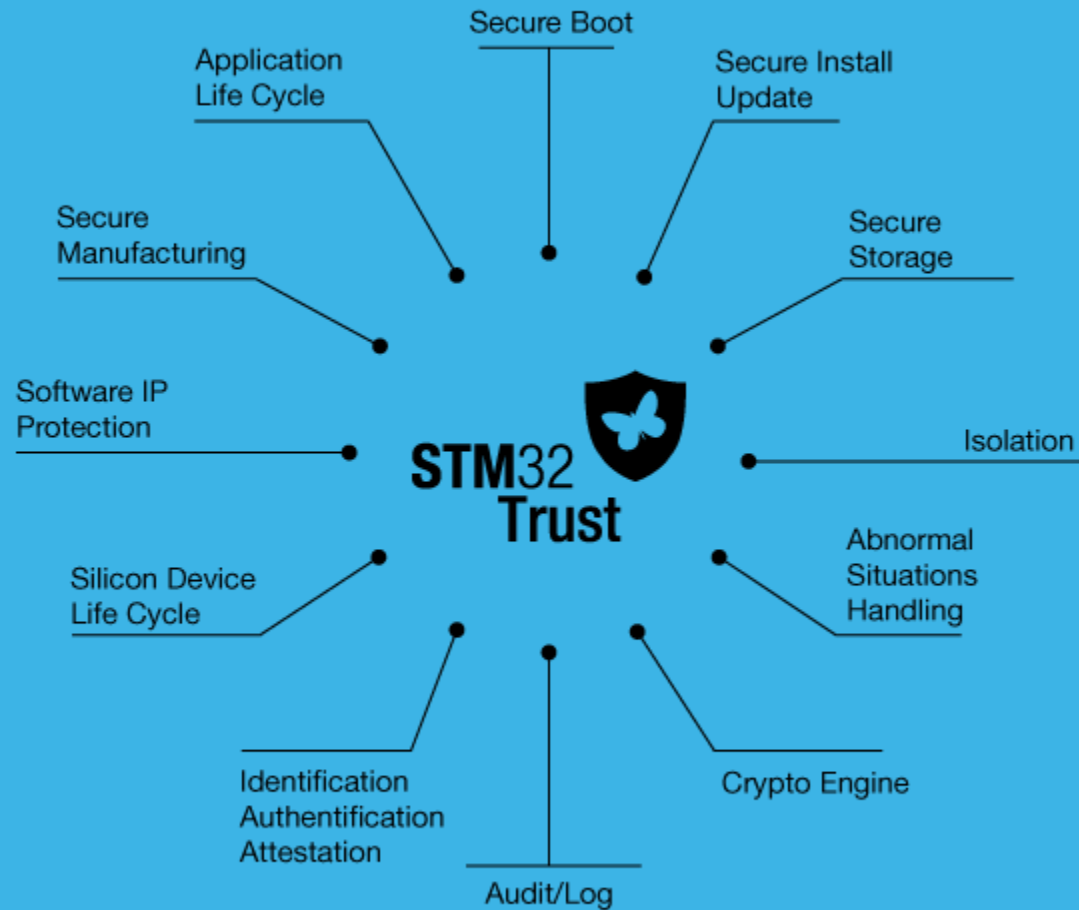


STM32H5 MCU



Target certifications


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STM32H5 MCU



Target certifications



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Our technology starts with You



Find out more at www.st.com/STM32Trust

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